

Material Behavior Prediction

Milestone 2

6 Figures Saved

Complete Technical Guide to Milestone 2: Exploratory Data Analysis (EDA)

1. Overview of Milestone 2 (EDA)

Exploratory Data Analysis is not about generating dozens of random plots. In machine learning for physical sciences, EDA serves two vital purposes:

1. **Statistical Validation:** Identifying distribution shapes, skewness, outliers, and collinearity to choose appropriate preprocessing and modeling algorithms.
2. **Physical Law Verification:** Confirming whether experimental observations obey known materials science principles (such as Abrams' Law and hydration kinetics).

2. The 6 Curated Visualizations & Scientific Answers

1 01_target_distribution.png • Distribution of Target (Strength)

Question Answered: How is concrete compressive strength distributed? Is it Gaussian (normal) or skewed?

Key Findings:

- Compressive strength ranges from **2.33 MPa** to **82.60 MPa** with a mean of **35.82 MPa** and median of **34.45 MPa**.
- The distribution is approximately bell-shaped with slight right-skew. Because $\text{mean} \approx \text{median}$, the target is well-behaved for regression models without requiring extreme log transformations.
- Civil engineering context: Standard structural concrete is 20–40 MPa; high-performance concrete exceeds 50 MPa.

2 02_feature_distributions.png • Feature Spread & Sparsity

Question Answered: What are the shapes and scales of our 8 input features?

Key Findings:

- **Sparsity:** Fly ash is 0 in **55.0%** of batches; Blast-furnace slag is 0 in **45.7%**; Superplasticizer is 0 in **36.8%**. These zeros are not missing data; they represent optional mix designs.
- **Skewness:** `age` is heavily right-skewed because testing concentrates at standard curing intervals (3, 7, 14, 28 days), with only a few long-term tests (90, 180, 365 days).

3 03_correlation_heatmap.png • Linear Feature-Target Correlations

Question Answered: Which ingredients linearly increase or decrease compressive strength?

Feature	Pearson r with Strength	Physical Interpretation
cement	+0.50	Strongest positive driver; more binder creates more crystalline load-bearing matrix.
age	+0.33	Strong positive driver; ongoing chemical hydration increases density over time.
superplastic	+0.37	Positive; enables low water content while maintaining fluidity.
water	-0.29	Negative; excess water evaporates, leaving microscopic voids that weaken structure.
fineagg / coarseagg	-0.17 / -0.16	Mild negative; excess aggregate reduces binder paste volume.

Multicollinearity Note: Superplasticizer and water have a strong negative correlation ($r = -0.66$). This matches chemistry: superplasticizer is an admixture designed to reduce required water volume.

4 04_water_cement_ratio_vs_strength.png • Abrams' Law Verification

Question Answered: Does the empirical dataset obey Abrams' Law (1918), the cornerstone law of concrete physics?

Key Findings:

- Abrams' Law states: $\text{Strength} = \frac{A}{B^{w/c}}$, meaning compressive strength falls exponentially as the water-to-cement ratio (w/c) increases.
- Our plot confirms a pronounced non-linear drop: as w/c increases from 0.3 to 1.2, strength drops from >70 MPa down to <20 MPa.
- Color-coding by curing age reveals that for any identical w/c ratio, older specimens (56–365 days) consistently exhibit higher strength due to prolonged hydration.

5 05_age_vs_strength.png • Curing Kinetics Over Time

Question Answered: How does material strength develop across time, and where does hydration plateau?

Key Findings:

- Strength development follows a steep logarithmic curve from Day 1 to Day 28.
- At 28 days (the standard structural testing milestone), concrete achieves the majority of its design strength.
- Hydration continues at a diminishing rate from 90 to 365 days, reaching asymptotic stability.

6 06_binder_type_vs_strength.png • Green Concrete & Supplementary Binders

Question Answered: Can sustainable industrial byproducts (slag & fly ash) match or exceed pure Portland cement?

Key Findings:

- Blended mixes (Cement + Slag) and Ternary mixes (Cement + Slag + Ash) achieve equal or higher median strength than pure Portland cement mixes.
- This proves the dataset contains high-performance, environmentally conscious formulations capable of achieving >50 MPa while significantly lowering the carbon footprint of construction materials.

3. Key Takeaways for Milestone 3 (Preprocessing)

Direct Guidance for Preprocessing & Modeling

1. **Feature Scaling is Essential:** Feature ranges vary by orders of magnitude (`age` : 1–365, `coarseagg` : 800–1150, `superplastic` : 0–32). Distance-sensitive models (Linear Regression, KNN, Ridge) require scaling (`StandardScaler`).
2. **Non-Linearity Requires Tree-Based Models:** The inverse relationship in Abrams' Law and the logarithmic hydration curve in curing age are non-linear. Linear regression will establish a baseline, but Decision Trees and Random Forests will capture these physical curves much more effectively.
3. **Zero Handling:** The zeros in slag, ash, and superplasticizer are physically meaningful (absence of admixture) and must be preserved as 0, not treated as missing values.